A Helmholtz Solver for Open Domains with Complex Geometry Using WaveHoltz and Overset Grids

Ngan Le

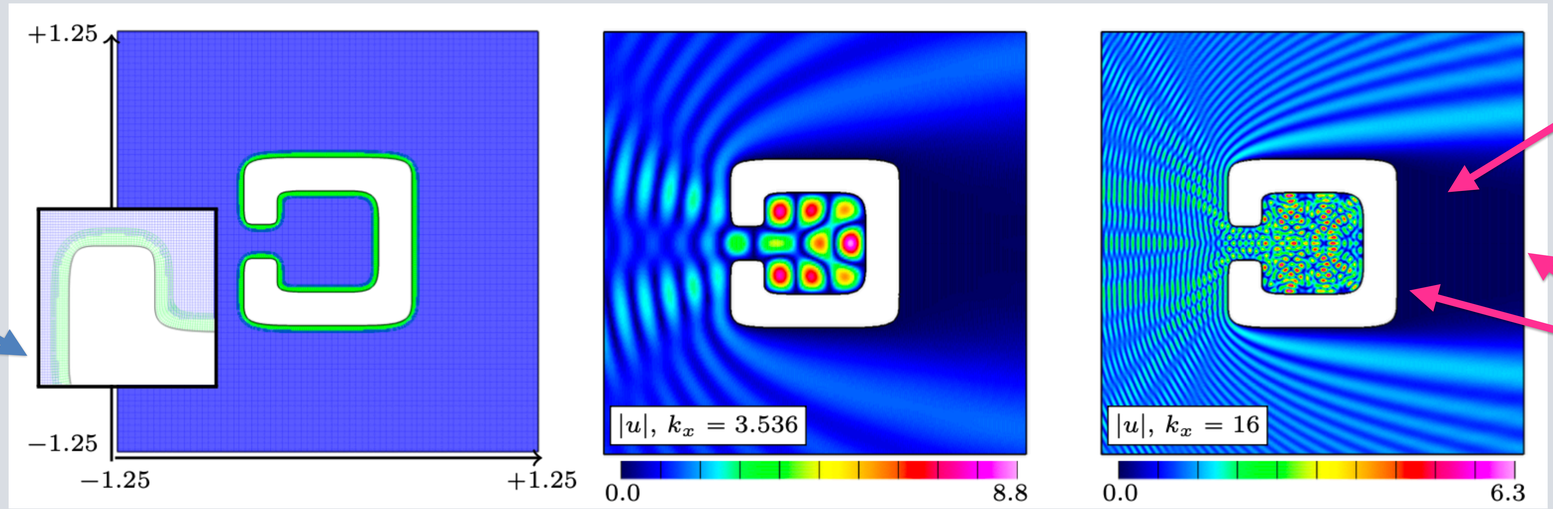
Department of Mathematical Sciences, Rensselaer Polytechnic Institute, Troy, NY, USA

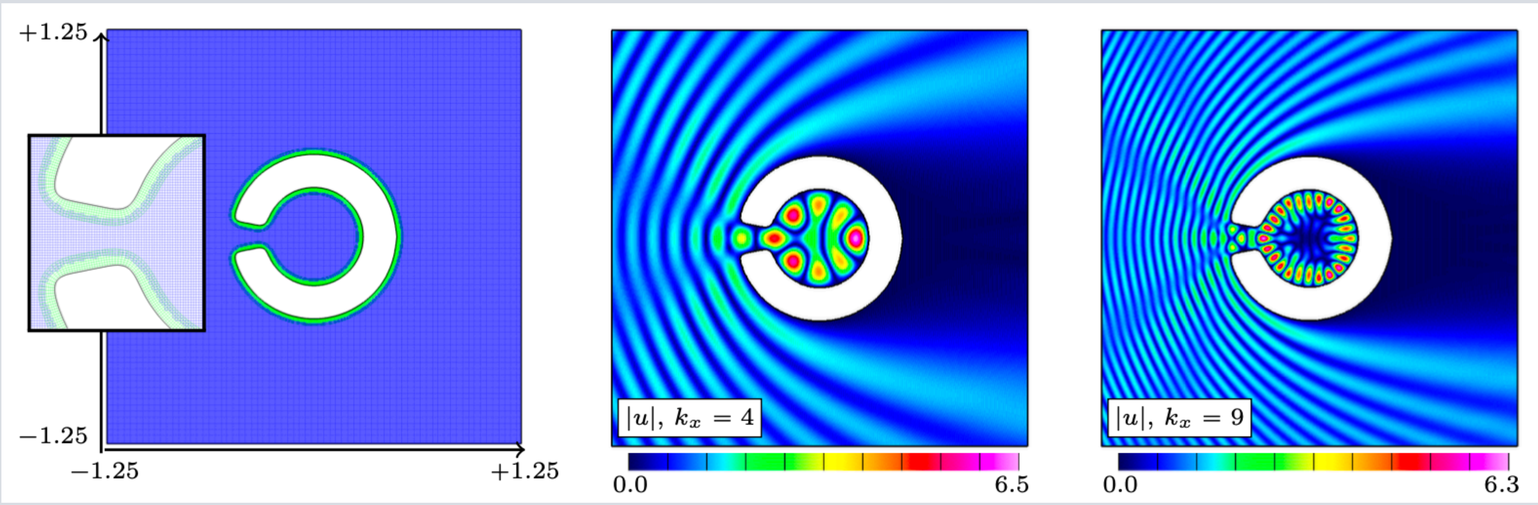
Collaborators:
- Jeffrey Banks, RPI
- William Henshaw, RPI
- Donald Schwendeman, RPI

Scattering of a plane wave from split-ring resonators

Sample of overset grid

Plane wave from the x direction





Plot of the total field

Radiation boundaries outside, Dirichlet inside

- Based on WaveHoltz algorithm
- Solve an associated wave equation
- Filter the solution in time
- Leads to a fixed-point iteration
- Can accelerate the convergence with Krylov-based iterative methods

Enables robust solutions of large-scale Helmholtz problems